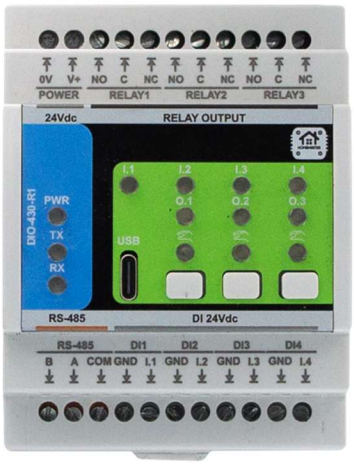



# Datasheet

## DIO-430-R1 Module

**Part No. DIO-430-R1**

**Manufacturer:** ISYSTEMS AUTOMATION S.R.L.
**Hardware Version:** V1.0
**Architecture:** Modular (MCU Board + Relay Board + Field Board)
**Controller:** ESP32-WROOM-32U

### Functional Overview

The system is implemented using three hardware boards:

| Board | Function |
|---|---|
| MCU Board | CPU, communication interfaces, USB configuration |
| Field Board | Digital inputs, signal protection, field interface |
| Relay Board | Relay outputs and load switching |

The DIO-430-R1 is a compact digital I/O module designed for industrial and building automation systems. It provides 4 isolated digital inputs and 3 relay outputs for monitoring field signals and controlling external loads. The module operates from a 24 V DC supply and includes input protection, relay switching capability, and communication interfaces for integration with PLCs and distributed control systems. Its modular architecture and DIN-rail form factor make it suitable for use in control cabinets for applications such as lighting control, HVAC automation, alarm monitoring, and general-purpose I/O expansion.

## Technical Specifications

| Specification | Details |
|---|---|
| Microcontroller | RP2350A dual-core microcontroller |
| Storage | External QSPI Flash (W25Q32JV) |
| Power Input | 24 V DC nominal (V+ / 0V) |
| Digital Inputs | 4 × isolated 24 V DC discrete inputs (DIx + GNDx), supporting dry-contact closure protected by per-channel PTC fuse, TVS surge suppression, and EMI filtering. |
| Relay Outputs | 3 × SPDT — 3 A MAX continuous per output (board/system limit).<br>Relay component contacts rated up to 12 A @ 250 V AC (resistive). |
| User Interface | 4 buttons, 13xLEDs(Power, 3 user LEDs, RX, TX, 3xRelay, 4xDI) |
| RS-485 | RS-485 Modbus RTU (MAX485, half-duplex) with TVS surge protection, PTC fuses, EMI choke, and fail-safe biasing |
| USB | USB-C (ESD protected, configuration and firmware upload) |
| File System | LittleFS persistent configuration storage |

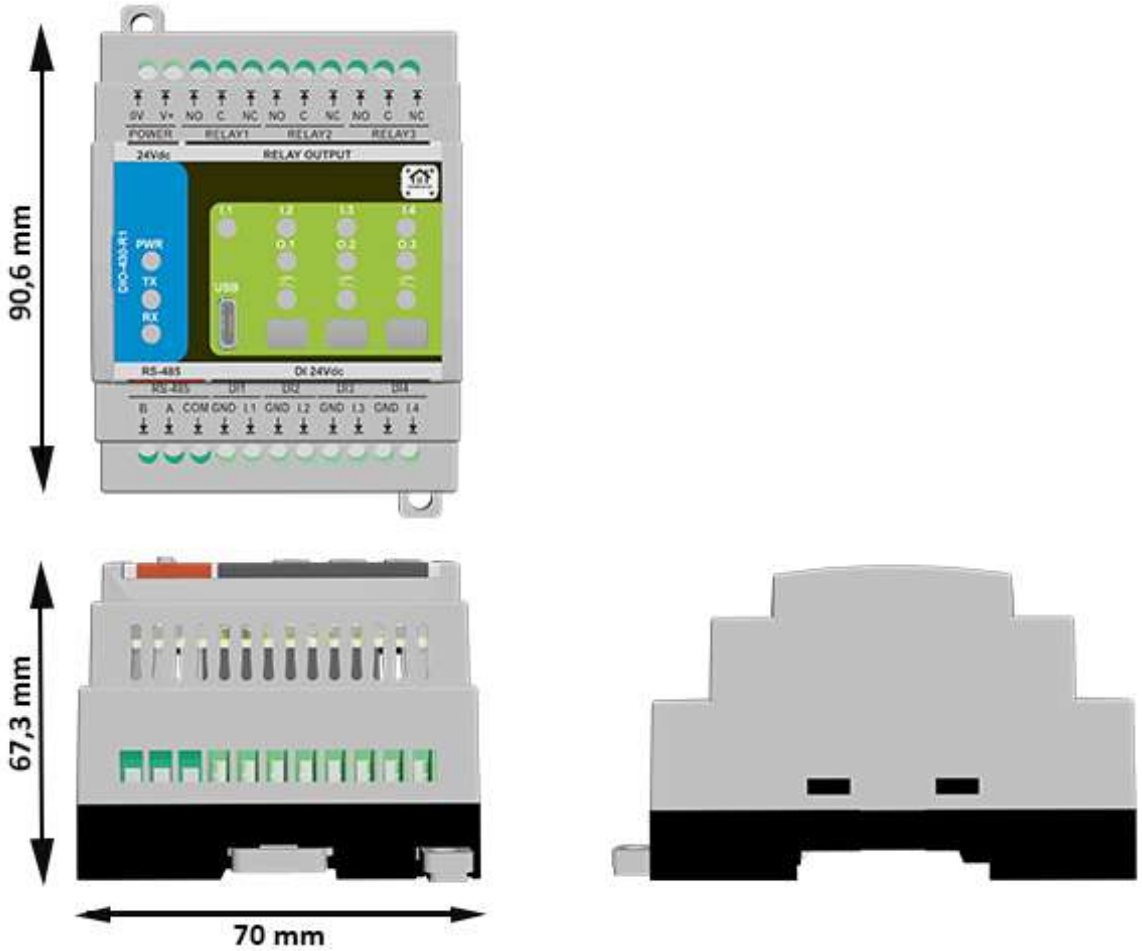
Note: Relay outputs are not internally fused. Use external overcurrent protection per local code and use an external contactor/relay for loads above 3 A or with high inrush/inductive characteristics.

## Installation, Environmental & Mechanical

| Category | Specification | Details |
|---|---|---|
| Terminal Specifications | Terminal type | Pluggable screw terminal blocks, 5.08 mm pitch |
| | Terminal pitch | 5.08 mm |
| | Wire cross-section | 0.2–2.5 mm² (AWG 24–12) |
| | Conductor type | Solid or stranded copper |
| | Stranded wire | Ferrules recommended |
| | Tightening torque | 0.4 Nm (max) |
| Environmental Ratings | Operating temperature | 0 °C to +40 °C |
| | Storage temperature | -10 °C to 55 °C |
| | Relative humidity | 0–90 % RH, non-condensing |
| | Ingress protection | IP20 (inside cabinet) |
| | Installation | Indoor control cabinet only; not for outdoor or exposed installation |
| | Terminal protection | All wiring terminals must be protected against accidental contact |
| Mechanical & Packaging | Product dimensions | 71.5 x 90 x 59 mm (LxWxH) |
| | DIN units | 4 division units (≈ 72 mm DIN rail mounting width) |
| | Mounting | 35 mm DIN rail |
| | Net weight | Xxx g |
| | Gross weight | Xxx+98 g |
| | Pack size | 140× 125 × 94 mm (L × W × H) |
| | Mechanical drawing | Front + side views with DIN-clip depth (see below) |
| | Notes | All dimensions shown in millimeters unless stated otherwise |

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

**All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.**



## Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

***General Routing Rules***

- Route low-level signal cables separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

***RS-485 Cable***

- **Construction:** twisted pair for **A/B**
- **Characteristic impedance: 120 Ω** (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples: J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**
- Use one twisted pair for **A/B**. Use the second pair for **COM (0V reference)** or spare.

***Shield Grounding***

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

## Compliance & Certifications

The HomeMaster MiniPLC is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

**EU Directives**

EMC 2014/30/EU · LVD 2014/35/EU · RED 2014/53/EU · RoHS 2011/65/EU

**Harmonised Standards**

| Area | Standards |
|---|---|
| EMC | EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions) |
| Electrical Safety | EN 62368-1 |
| RoHS | EN IEC 63000 |

**Radio**

The product integrates a pre-certified ESP32 Wi-Fi radio module (2.4 GHz). Final product conformity with the Radio Equipment Directive is demonstrated by the maintained technical documentation and conformity assessment of the complete device.

**Safety notice**

L / N: hazardous voltage · 24 V DC: SELV · Qualified personnel only

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